Hybrid Sphere Decoding with Embedded Initialization and Explicit Pruning

Core Mathematical Argument and Controlled Experimental Evidence

This concise report retains only the evidence required to justify the proposed hybrid SDA: complete-sequence equivalence, embedded upper-bound quality, explicit optimality-certification efficiency, and total hybrid search effort. Exploratory diagnostics are maintained separately in the detailed analysis report and are intentionally omitted here.

October 2026

Contents

1 Research Question and Claim **1**

2 Claim 1: Complete-Sequence Equivalence **2**

3 Claim 2: Embedded Initialization Provides a Better Upper Bound **4**

4 Claim 3: Explicit Search Certifies Optimality More Efficiently **6**

4.1 Secondary analysis of the Cholesky-factor distribution 7

5 Hybrid Algorithm and Correctness **10**

6 Controlled Ablation Result **11**

6.1 Dataset, pairing, and sample selection 11

6.2 Meaning of the five configurations 11

6.3 Replay validation 13

7 Closed-Loop Deployment Validation **15**

7.1 Time-domain overview 15

8 Conclusion and Scope **20**

1 Research Question and Claim

The controller optimizes a binary switching sequence

$$\boldsymbol{U} = [S_{a,1}, S_{b,1}, S_{c,1}, \ldots, S_{a,N_p}, S_{b,N_p}, S_{c,N_p}]^{\mathsf{T}}, \qquad \boldsymbol{U} \in \{0,1\}^{3N_p}, \tag{1.1}$$

with $N_p = 5$. Two representations of the zero-vector objective are available. The *explicit* formulation adds λ_z when a completed three-phase switching vector is 000 or 111. The *embedded* formulation absorbs an equivalent quadratic expression into the matrix factor used by the sphere decoder.

The proposed method does not select one representation for the entire solver. It uses their complementary properties:

$$\boxed{\text{embedded Babai initialization} + \text{ explicit branch-and-bound search.}} \tag{1.2}$$

The academic argument consists of three claims:

1. complete-sequence equivalence permits a candidate to be transferred safely between the formulations;
2. the embedded representation provides a better initial feasible upper bound;
3. the explicit representation certifies optimality with fewer nodes once initializer quality is controlled.

Terminology. An *incumbent* is the best complete feasible sequence found so far; its cost is the current upper bound or sphere radius R. A *prefix* is an incomplete sequence containing the first i binary decisions. Its partial metric P_i is a lower bound used for pruning. A node is pruned when $P_i > R$. Thus upper-bound quality and prefix-bound quality are distinct properties.

2 Claim 1: Complete-Sequence Equivalence

For one three-phase vector define

$$m = S_a + S_b + S_c, \qquad m \in \{0, 1, 2, 3\}. \tag{2.1}$$

The explicit zero-vector cost is represented exactly on the binary domain by

$$J_{\text{z,exp}}(m) = \frac{\lambda_z}{2}(m-1)(m-2). \tag{2.2}$$

The embedded centered quadratic is

$$J_{\text{z,emb}}(m) = \frac{\lambda_z}{2}(m-1.5)^2. \tag{2.3}$$

Expanding both expressions gives

$$J_{\text{z,emb}}(m) = J_{\text{z,exp}}(m) + \frac{\lambda_z}{8}. \tag{2.4}$$

Proposition 2.1 (Leaf-level equivalence)**.** *For every complete binary sequence,*

$$J_{\text{emb}}(\boldsymbol{U}) = J_{\text{exp}}(\boldsymbol{U}) + C, \qquad C = \frac{N_p \lambda_z}{8}. \tag{2.5}$$

Consequently, both formulations have the same minimizer set.

Proof. The per-block difference is independent of m. Summation over N_p blocks adds the same constant to every feasible complete sequence and therefore preserves every objective ordering. □

For $N_p = 5$ and $\lambda_z = 10{,}000$,

$$C = 6250. \tag{2.6}$$

The numerical evidence is deliberately minimal but exhaustive at the selected analysis points: all 2^{15} complete sequences exhibited the same difference 6249.99992 to floating-point accuracy. All paired closed-loop samples also returned identical optimal switching

sequences. This establishes that an embedded Babai sequence is a valid candidate for the explicit problem once its cost is reevaluated under the explicit objective.

3 Claim 2: Embedded Initialization Provides a Better Upper Bound

Let the initial optimality gap be

$$G_0 = R_0 - J^\star \geq 0, \tag{3.1}$$

where R_0 is the feasible initial incumbent obtained by evaluating the initializer's switching sequence under the objective being searched, and $J^\star$ is the globally optimal cost in the corresponding representation. A smaller G_0 means a better initial feasible upper bound; $G_0 = 0$ means that the initializer already supplies an optimal sequence.

The native explicit implementation selects Round, Shift, or Babai using only the quadratic tracking distance and adds the explicit zero-vector/dead-time cost afterward. Consequently, a sequence that is geometrically close to the unconstrained center can still receive a large discrete penalty. The embedded Babai construction already includes the zero-vector term in its geometry.

Table 3.1: Initial upper-bound quality over 500 transactions per torque.

	Native explicit		Embedded Babai	
Torque	Median G_0	Optimal at start	Median G_0	Optimal at start
10 Nm	49279	0%	≈ 0	51.8%
50 Nm	39947	0%	2.55	49.2%
90 Nm	39879	0%	57.12	40.0%

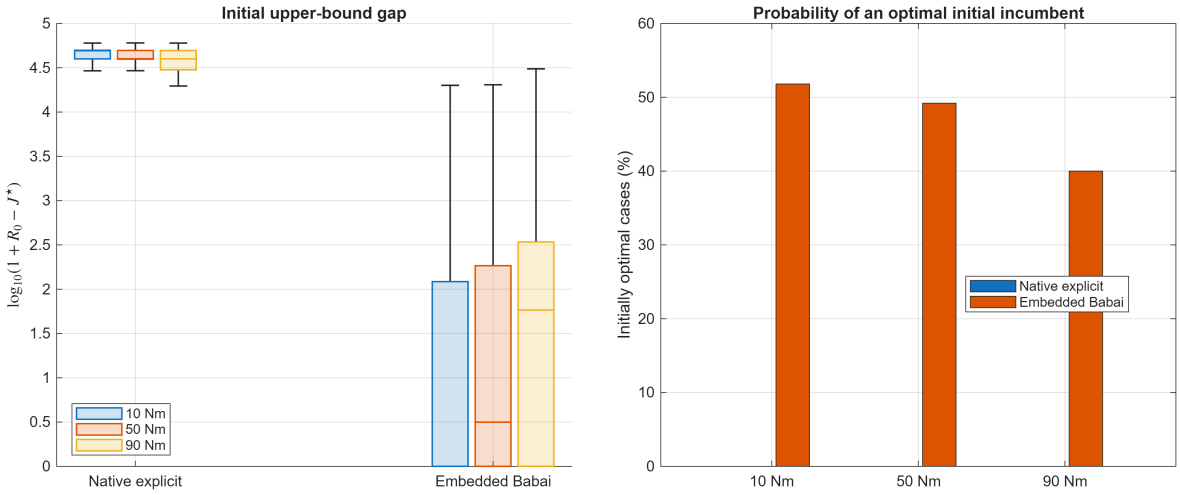


Figure 3.1: Direct evidence for the embedded upper-bound claim. Left: initial optimality gap from equation (3.1), displayed as $\log_{10}(1 + G_0)$. Right: percentage of transactions in which the initial candidate is already globally optimal.

The result supports a scoped empirical statement: under the investigated horizon, weight, variable ordering, and operating conditions, the embedded Babai construction gives a substantially tighter initial upper bound. It does not claim that embedded Babai is optimal for every possible state.

4 Claim 3: Explicit Search Certifies Optimality More Efficiently

At depth $i = 3q$, exactly q complete three-phase switching vectors are known. For the implemented forward ordering, the same-prefix metrics satisfy

$$P_{3q}^{\mathrm{emb}} - P_{3q}^{\mathrm{exp}} = q\frac{\lambda_z}{8}. \tag{4.1}$$

After normalizing the complete embedded objective by $C = N_p\lambda_z/8$,

$$P_{3q}^{\mathrm{exp}} - \widetilde{P}_{3q}^{\mathrm{emb}} = (N_p - q)\frac{\lambda_z}{8} > 0, \qquad q < N_p. \tag{4.2}$$

Thus the explicit metric is tighter at the completed switching-vector boundaries. Numerically, the accumulated offsets at depths $3, 6, 9, 12, 15$ are exactly $1250, 2500, 3750, 5000, 6250$ for all enumerated prefixes.

The strongest experimental isolation is the oracle comparison. Both searches are initialized with the already known optimal sequence. Initialization quality is therefore identical and cannot explain the remaining node-count difference.

Table 4.1: Mean nodes required when the globally optimal incumbent is supplied before search.

Method	10 Nm	50 Nm	90 Nm
Oracle explicit	160.49	161.12	150.97
Oracle embedded	1123.00	1137.90	1010.40
Explicit reduction	85.7%	85.8%	85.1%

The explicit search therefore requires only approximately 14–15% of the embedded node count to certify the same already known optimum. Combined with equation (4.2), this provides direct mathematical and controlled experimental evidence for the explicit proof-stage advantage. The statement is scoped to the implemented factorization and variable ordering; universal node dominance at every intermediate depth is not claimed.

4.1 Secondary analysis of the Cholesky-factor distribution

The recursive increment can be written in terms of the conditional center

$$c_i = \frac{(HU_{\mathrm{unc}})_i - H_{i,1:i-1}U_{1:i-1}}{H_{ii}} \tag{4.3}$$

as

$$\Delta P_i = H_{ii}^2 (U_i - c_i)^2. \tag{4.4}$$

Thus, the row norm alone is not the level-wise discrimination gain. The diagonal magnitude $|H_{ii}|$ scales the present binary decision, while the preceding row coefficients determine how strongly the center depends on the already selected prefix. To separate these effects, define

$$\kappa_i = \frac{\|H_{i,1:i-1}\|_2}{|H_{ii}|}. \tag{4.5}$$

The row norm, diagonal magnitude, and coupling ratio were evaluated on the same 1500 transactions used by the oracle experiment. Their formulation differences were correlated with

$$\Delta N_{\mathrm{oracle}} = N_{\mathrm{oracle,emb}} - N_{\mathrm{oracle,exp}} \tag{4.6}$$

using Spearman rank correlation.

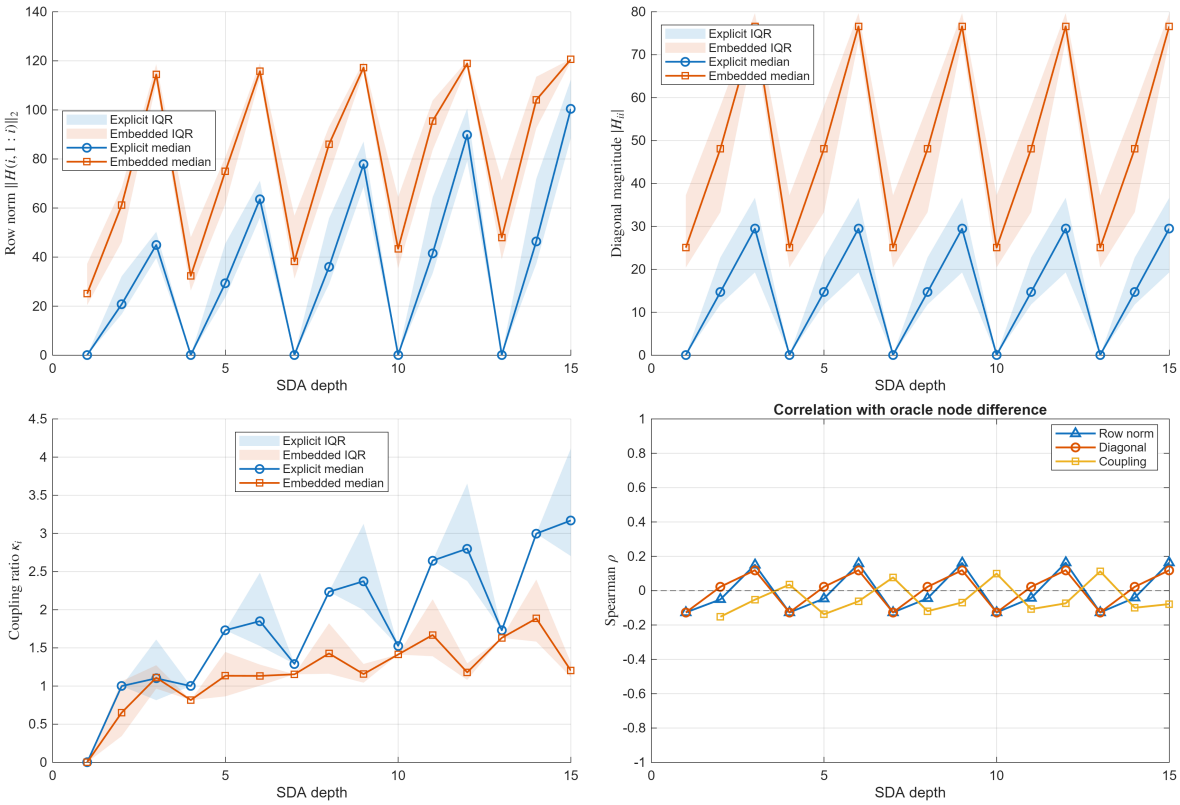


Figure 4.1: Cholesky-factor structure under a common optimal incumbent. Shaded regions are interquartile ranges over all sampled transactions. The lower right panel correlates the embedded–explicit difference of each structural quantity with the paired oracle node-count difference.

Table 4.2: Largest absolute depth-wise Spearman correlation with the oracle node difference. Small p-values reflect the sample count and do not imply a large effect size.

Structural quantity	Maximum $\lvert\rho\rvert$	Depth	Interpretation
Row norm difference	0.164	15	weak association
Diagonal difference	0.127	repeated block positions	weak association
Coupling-ratio difference	0.152	2	weak association

The embedded factor clearly redistributes level-wise scaling: it has larger row norms and diagonal magnitudes, while the explicit factor develops larger normalized coupling at many deeper levels. Nevertheless, all observed correlations satisfy $|\rho| < 0.17$. These simple marginal descriptors therefore do not explain the order-of-magnitude oracle node difference on their own. Their influence acts jointly through the path-dependent centers

in equation (4.4), enumeration order, and accumulated prefix bounds. This result rules out the overly simple explanation that the embedded search is slower merely because its rows are “flatter” or larger. The primary evidence remains the block-boundary prefix relation and the controlled oracle experiment.

5 Hybrid Algorithm and Correctness

The hybrid first computes

$$\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}} = \mathrm{Babai}(\boldsymbol{H}_{\mathrm{emb}}, \boldsymbol{U}_{\mathrm{unc,emb}}). \tag{5.1}$$

It then reevaluates this sequence under the explicit objective:

$$R_{0,\mathrm{hyb}} = J_{\mathrm{exp}}(\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}}), \tag{5.2}$$

and starts the original explicit SDA using this sequence and radius.

Proposition 5.1 (Global optimality preservation)**.** *If the explicit SDA terminates without reaching its node limit, replacing its native initial candidate by the embedded Babai candidate does not change the globally optimal output.*

Proof. $\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}}$ is a complete feasible binary sequence, so its explicit cost is an upper bound on the explicit optimum. The subsequent explicit branch-and-bound algorithm and all of its valid lower bounds are unchanged. The initializer can change the search order and pruning amount but cannot remove a branch capable of improving the incumbent. □

6 Controlled Ablation Result

6.1 Dataset, pairing, and sample selection

All results in this chapter originate from the same SDI archive, `embedded_H.mldatx`. The archive contains 40,000 SDA transactions for each of the following six closed-loop runs:

$$\begin{aligned} &\texttt{original_10NM}, \quad \texttt{original_50NM}, \quad \texttt{original_90NM}, \\ &\texttt{embedded_H_10NM}, \quad \texttt{embedded_H_50NM}, \quad \texttt{embedded_H_90NM}. \end{aligned} \tag{6.1}$$

The auxiliary run `original_no_zero_lambda_10NM` is not used in the hybrid or oracle comparison.

The interval before $t = 0.05$ s is excluded to avoid the initial startup transient. From the remaining transactions, 500 indices are selected at uniformly spaced positions for each torque condition. The same indices are used for the original and embedded runs, producing

$$3 \text{ torque conditions} \times 500 = 1500 \tag{6.2}$$

paired physical operating points. "Paired" means that the two formulations are compared at the same torque label, controller time, and transaction index. The selection is deterministic and is not based on the resulting node count.

For every selected transaction, the replay input contains the logged factor H, unconstrained center U_{unc}, previous switching sequence, compensated current, current-increment matrices, node limit, zero-vector and dead-time weights, dead-time ratio, and current-sign margin. Each formulation retains its own logged H and U_{unc}; only the physical transaction index is paired.

6.2 Meaning of the five configurations

Table 6.1 states exactly which data and initializer are used by each configuration.

Table 6.1: Inputs and initialization used in the ablation study.

Configuration	Matrix and center	Initial complete sequence	Purpose
Native explicit	From the paired `original` run	Best quadratic Round/Shift/Babai candidate selected by the production logic	Production reference
Native embedded	From the paired `embedded_H` run	Best quadratic Round/Shift/Babai candidate in embedded geometry	Embedded reference
Hybrid	Explicit H and center for search; embedded H and center only for Babai	Embedded Babai sequence, reevaluated under the explicit objective	Proposed implementable method
Oracle explicit	From the paired `original` run	Logged globally optimal sequence $U^\star$	Isolate explicit certification effort
Oracle embedded	From the paired `embedded_H` run	The same logged globally optimal binary sequence $U^\star$	Isolate embedded certification effort

The oracle configurations do not constitute a separate dataset. They replay the same 1500 paired transactions but replace the practical initializer by the already known logged optimum. Each oracle recomputes the cost of $U^\star$ under its own objective:

$$R_{0,\text{oracle,exp}} = J_{\text{exp}}(U^\star), \qquad R_{0,\text{oracle,emb}} = J_{\text{emb}}(U^\star). \tag{6.3}$$

The numerical radii therefore differ by 6250; the shared quantity is the optimal binary sequence, not the raw radius. Oracle results are used only to remove initializer quality as a confounding variable. Because they use the known solution, they are diagnostic lower baselines and are not presented as deployable controllers.

6.3 Replay validation

Every replayed configuration is accepted only if its final cost and switching sequence agree with the logged reference within numerical tolerance. Thus the reported node counts compare alternative search procedures on the same solved optimization instances rather than different control trajectories. The closed-loop plant is not resimulated during this offline analysis.

Five configurations were evaluated on the same 500 transactions per torque: native explicit, native embedded, the implementable hybrid, and two diagnostic oracle searches. Every configuration reproduced the logged optimal sequence and cost.

Table 6.2: Total search effort. Entries are mean nodes with Q95 in parentheses.

Method	10 Nm	50 Nm	90 Nm
Native explicit	451.96 (919)	466.67 (1033)	442.48 (867)
Native embedded	1140.90 (1794)	1179.00 (1845)	1100.20 (1745)
Hybrid	**182.66 (489)**	**191.49 (500)**	**199.92 (504)**
Oracle explicit	160.49 (427)	161.12 (380)	150.97 (313)
Oracle embedded	1123.00 (1746)	1137.90 (1808)	1010.40 (1662)

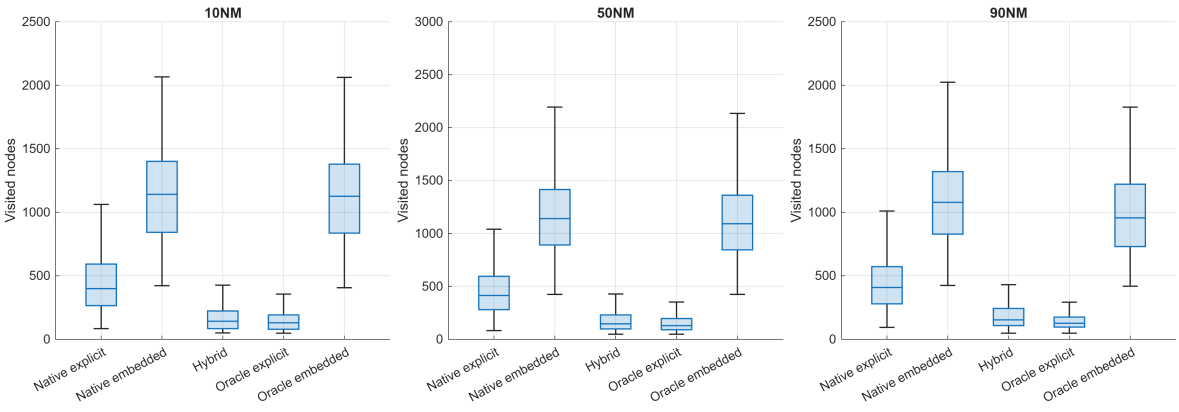


Figure 6.1: Node-count distributions for the complete ablation. The hybrid is the lowest-node implementable configuration and approaches the diagnostic oracle-explicit baseline.

Relative to the native explicit solver, the hybrid reduces mean nodes by 59.6%, 59.0%, and 54.8% at 10, 50, and 90 Nm. Relative to native embedded, the reductions are 84.0%,

83.8%, and 81.8%. These reductions are obtained without changing the returned globally optimal switching sequence.

7 Closed-Loop Deployment Validation

The offline replay isolates the search mechanism, but it does not by itself show that the complete Simulink feedback loop preserves the same behavior. The hybrid solver was therefore deployed in `FCSMPC_double_vector_2025b`. Its 10-kHz stage forms both factors, its 40-kHz stage forms both matched unconstrained centers after one current reconstruction, and its SDA stage uses the embedded pair only for Babai before returning to the explicit pair for search. The original and embedded-only paths were commented out during the hybrid runs.

The archive `hybrid_SDA.mldatx` contains one-second runs at 10, 50, and 90 Nm. After excluding $t < 0.05$ s, each condition contributes 38,001 transactions. These transactions were paired by timestamp with the original and embedded runs in `embedded_H.mldatx`, giving 114,003 comparisons. All methods returned `optimal=true`, and none reached the 10,000-node limit.

7.1 Time-domain overview

Figures 7.1–7.3 provide a direct view of the three node-count signals before they are reduced to statistical summaries. Since plotting every 40-kHz optimizer call would form an opaque band, the one-second record is divided into non-overlapping 1-ms bins. Each displayed point is the arithmetic mean of the approximately 40 synchronized transactions in that bin. This operation is for visualization only; the tables and distributional results retain every transaction after 50 ms.

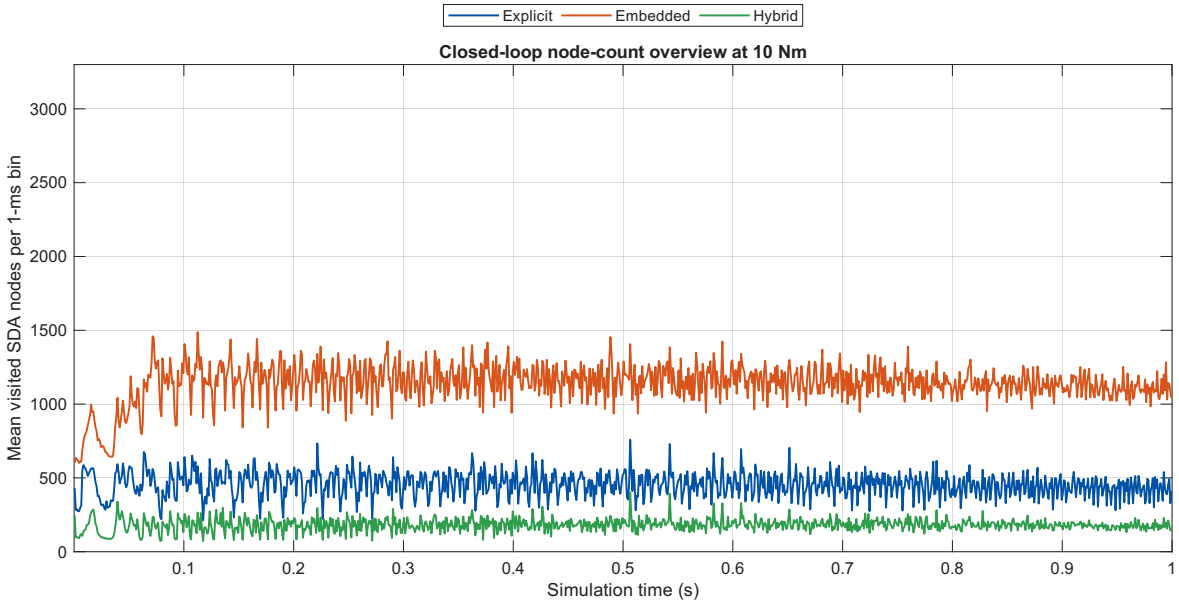


Figure 7.1: Visited-node time series at 10 Nm, displayed as 1-ms bin means over the complete simulation.

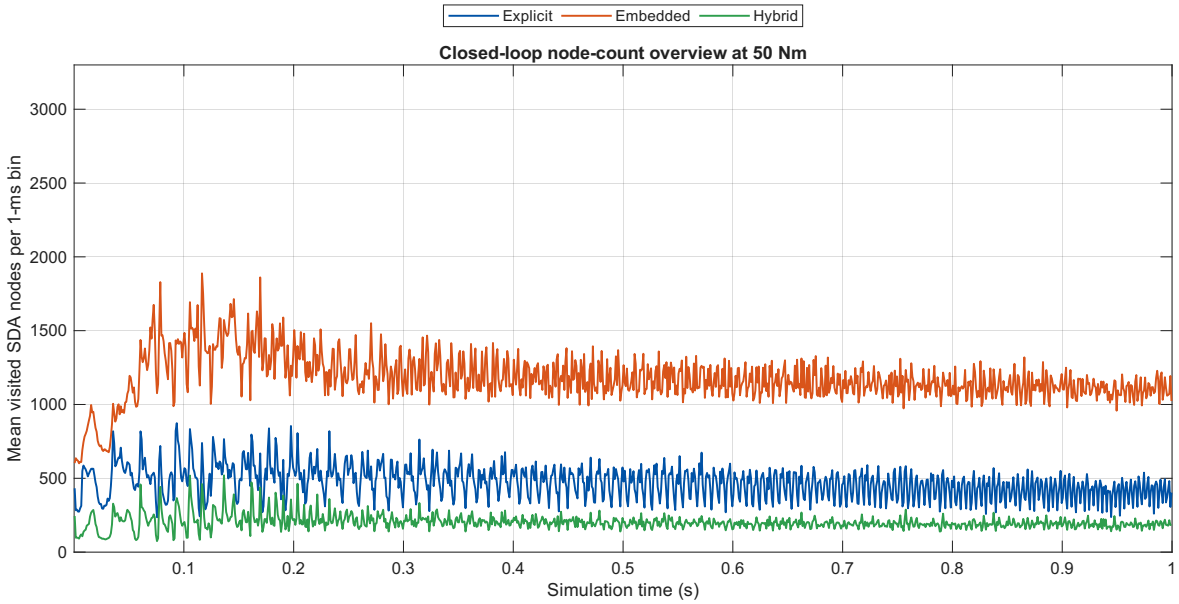


Figure 7.2: Visited-node time series at 50 Nm, displayed as 1-ms bin means over the complete simulation.

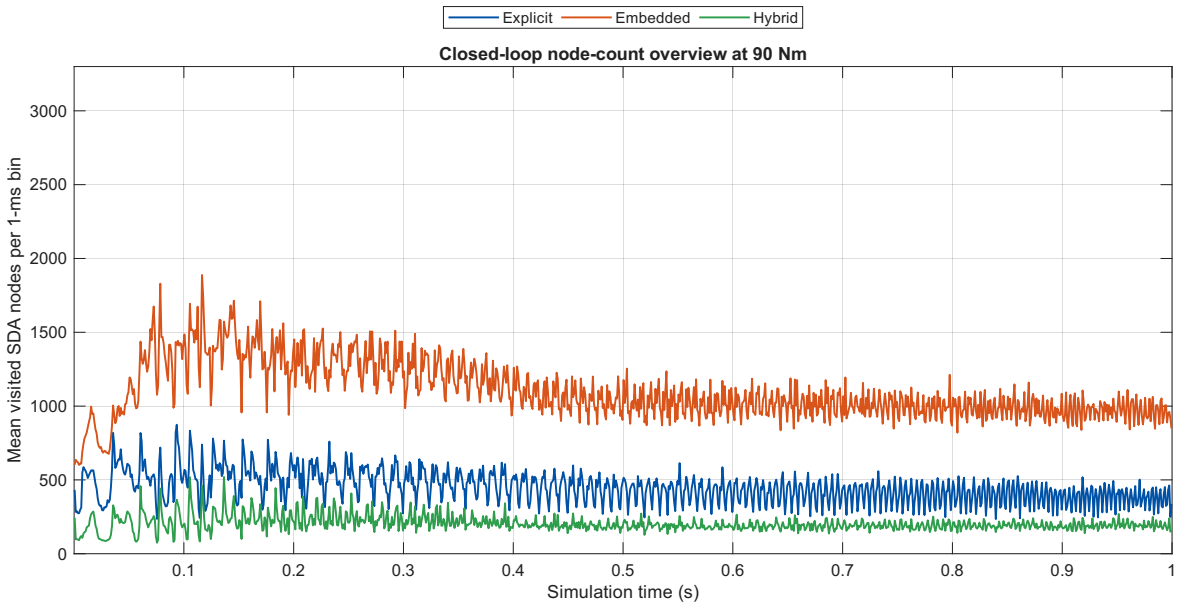


Figure 7.3: Visited-node time series at 90 Nm, displayed as 1-ms bin means over the complete simulation.

Across all three loads, embedded-only search is generally the most expensive, explicit search is intermediate, and the hybrid is the least expensive. The initial transient remains visible in these overview plots but is excluded from the steady-state statistics below.

Table 7.1: Deployed closed-loop search effort over all steady-state transactions.

Torque	Method	Mean	Median	Q95	Maximum
10 Nm	Explicit	456.16	402	930.0	2318
	Embedded	1157.50	1162	1788.9	3110
	Hybrid	**182.63**	**138**	**498.0**	**1678**
50 Nm	Explicit	469.87	416	998.0	2134
	Embedded	1197.10	1160	1872.0	3116
	Hybrid	**204.14**	**158**	**500.0**	**1732**
90 Nm	Explicit	446.87	402	906.9	2030
	Embedded	1115.40	1088	1764.0	3116
	Hybrid	**205.57**	**160**	**492.0**	**1696**

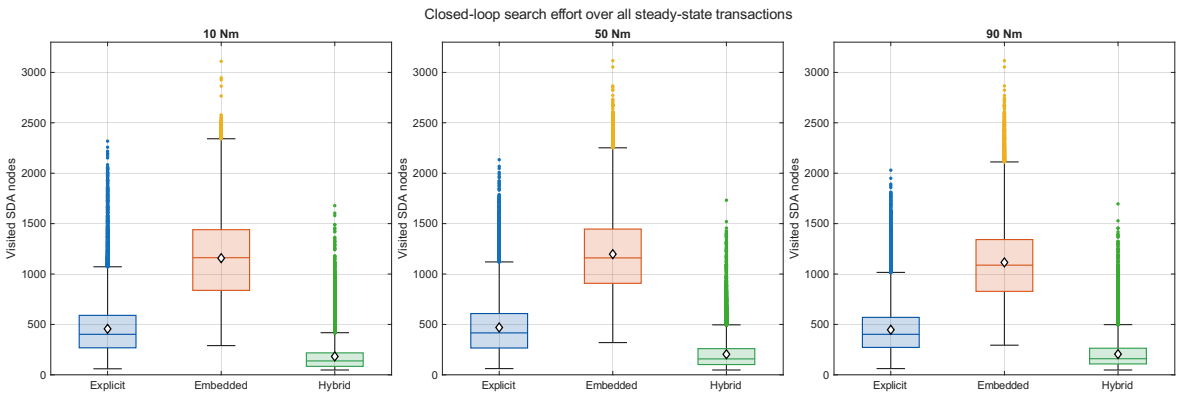


Figure 7.4: Closed-loop node-count distributions for all retained transactions.

The mean reductions relative to native explicit are 59.96%, 56.55%, and 54.00%; relative to embedded-only they are 84.22%, 82.95%, and 81.57%. The hybrid and both reference formulations select identical optimal sequences for every paired transaction. The hybrid–explicit cost difference is below 1.9×10^{-12}, while the median embedded–hybrid difference is exactly 6250 at all three loads.

Table 7.2: Embedded-Babai upper-bound quality in closed-loop deployment.

Torque	Initially optimal	Median G_0	Reduction vs explicit	Reduction vs embedded
10 Nm	52.26%	3.4×10^{-13}	59.96%	84.22%
50 Nm	43.07%	36.38	56.55%	82.95%
90 Nm	39.37%	67.82	54.00%	81.57%

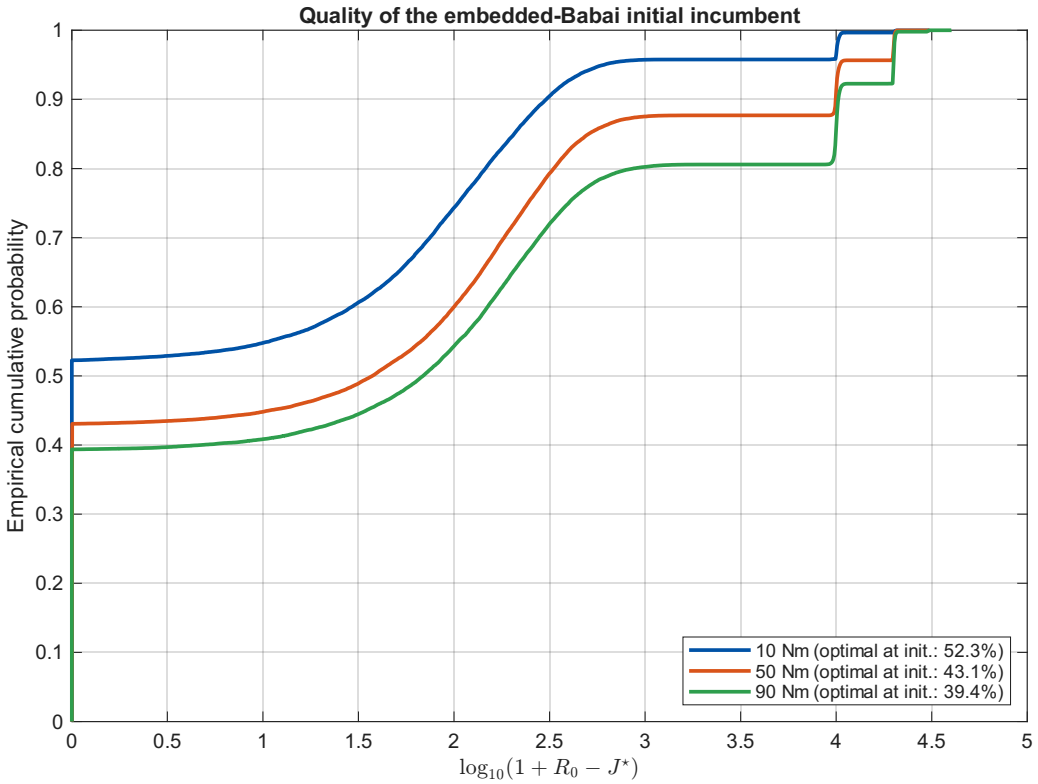


Figure 7.5: Empirical distribution of the embedded-Babai initial explicit-cost gap in the deployed closed-loop controller.

A point (x, y) means that the empirical fraction y of transactions obeys $R_0 - J^\star \le 10^x - 1$. This follows from the plotted transformation

$$X = \log_{10}(1 + G_0), \qquad G_0 = 10^X - 1, \tag{7.1}$$

and its empirical cumulative distribution

$$\widehat{F}_X(x) = \frac{1}{N} \sum_{k=1}^{N} \mathbf{1}\{\log_{10}(1 + G_{0,k}) \le x\}. \tag{7.2}$$

Thus, $x = 0, 1, 2, 3$ represents $G_0 = 0, 9, 99, 999$, respectively. At a fixed cumulative probability, the curve with the smaller horizontal coordinate represents the smaller initial gap and hence the tighter initial incumbent. Equivalently, at a fixed gap threshold, the higher empirical CDF indicates better initialization quality. The jump at $x = 0$ equals the fraction with zero initial gap: embedded Babai has already supplied the globally optimal sequence in those transactions, although the explicit tree must still certify it.

The deployed results therefore reproduce the offline mechanism: embedded Babai supplies a tight incumbent, while explicit prefix bounds complete the proof with fewer nodes. They establish node-count reduction, not yet an equal processor-time reduction.

8 Conclusion and Scope

The evidence supports the following chain:

$$\boxed{\begin{array}{c}\text{complete-sequence equivalence}\\ \Downarrow\\ \text{embedded Babai supplies a tighter feasible upper bound}\\ \Downarrow\\ \text{explicit branch penalties supply more effective certification}\\ \Downarrow\\ \text{the hybrid preserves optimality and reduces visited nodes.}\end{array}} \tag{8.1}$$

The present result establishes search-node reduction, not yet hardware-time reduction. Final engineering validation must include the additional embedded factor construction, Babai pass, memory use, mean execution time, Q99 execution time, and worst-case execution time relative to the 10/40 kHz deadlines.

Exploratory KDEs, pruning-margin plots, dynamic-radius traces, representative Q5/Q50/Q95 cases, and full same-prefix distributions remain available in the detailed analysis report but are not required for the core argument.